

STUDENT PERFORMANCE FACTORS ANALYSIS

*A Comprehensive Statistical Report on Academic Performance
Determinants*

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Dataset: Student Performance Factors (Kaggle)

Sample Size: N = 6,607 students

Number of Variables: 20

Missing Values: None (0 NA across all variables)

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1. INTRODUCTION AND STUDY OVERVIEW

This report presents a comprehensive statistical analysis of student performance factors using a dataset comprising 6,607 student records with 20 variables. The primary objective is to identify and quantify the key determinants of academic success, as measured by exam scores. The dataset is complete with no missing values, ensuring the reliability of all statistical inferences.

Data Types:

- Numerical variables: Study hours, attendance, exam scores, sleep hours, previous scores, tutoring sessions, physical activity
- Categorical variables: Gender, motivation level, school type, family income, internet access, teacher quality

The analysis employs multiple statistical techniques including descriptive statistics, correlation analysis, linear regression, Lasso regression for variable selection, and cluster analysis to provide a multi-dimensional understanding of student performance drivers.

2. DESCRIPTIVE STATISTICS

Table 1 presents the central tendency and dispersion measures for key continuous variables. The descriptive statistics indicate that students achieve an average exam score of 67.24 with relatively low variability ($SD = 3.89$), suggesting consistent performance across the sample. Attendance and study hours show moderate variation, while sleep hours remain relatively stable. Overall, the data suggest that students maintain similar academic performance levels, with some variation in effort-related factors such as study time and attendance.

Table 1: Descriptive Statistics

Variable	Mean	SD	Min	Max
Exam Score	67.24	3.89	55	101
Hours Studied	19.98	5.99	1	44
Attendance	79.98	11.55	60	100
Sleep Hours	7.03	1.47	4	10
Previous Scores	75.07	14.40	50	100

Table 2: Extended Descriptive Statistics (with Skewness and Kurtosis)

Variable	Mean	SD	Median	Min	Max	Skew	Kurtosis
Hours Studied	19.98	5.99	20	1	44	0.01	0.02
Attendance	79.98	11.55	80	60	100	0.01	-1.19
Sleep Hours	7.03	1.47	7	4	10	-0.02	-0.50
Previous Scores	75.07	14.40	75	50	100	0.00	-1.19
Tutoring Sessions	1.49	1.23	1	0	8	0.82	0.64
Physical Activity	2.97	1.03	3	0	6	-0.03	-0.06
Exam Score	67.24	3.89	67	55	101	1.64	10.56

Key Insights from Descriptive Statistics:

- Average study time is approximately 20 hours, with attendance at ~80% and exam scores averaging ~67.
- The exam score distribution shows high kurtosis (10.56) and positive skewness (1.64), indicating a concentration of scores around the mean with a long right tail of high achievers.
- Tutoring sessions display moderate positive skewness (0.82), suggesting most students have few tutoring sessions with some having many.
- The dataset is clean and reliable with no missing values, making it suitable for robust statistical analysis.

3. DATA DISTRIBUTION AND OUTLIER ANALYSIS

The exam scores are mostly concentrated around 65–70, showing average student performance. The distribution is slightly right-skewed, meaning a few students scored much higher than the majority. Because exam score is the main outcome (dependent variable), its distribution warrants careful examination before proceeding with inferential analyses.

Outlier Detection (IQR Method):

- $Q1 = 65$, $Q3 = 69$, $IQR = 4$
- Lower fence = 59, Upper fence = 75
- Scores > 75 flagged as outliers

Interpretation: The majority of students cluster tightly around 65–69; high achievers scoring 75+ represent statistically exceptional performance, warranting investigation into what drives these outliers.

Table 3: Exam Score Distribution (Binned Analysis)

Score Range	Frequency	Percentage
50–60	145	2.2%
60–70	5,379	81.4%
70–80	1,040	15.7%
80+	42	0.6%
Total	6,607	100.0%

Key Takeaway: The distribution is severely compressed within the pass range, with minimal score differentiation. The vast majority of students (81.4%) score between 60–70, confirming the exam score distribution is heavily concentrated around average performance with very few low or high achievers. High performers (80+) are extremely rare at only 0.6%. This compression may limit predictive power of regression models and suggests investigating whether the exam lacks discriminatory power at the upper end.

Gender Comparison:

Based on the box plot analysis of exam scores by gender, central tendency is nearly identical: both female and male students have the same median exam score (approximately 67) and very similar interquartile ranges, indicating that gender has no meaningful effect on typical exam performance. Males show slightly more score dispersion at the upper end with outliers extending to ~100, suggesting males may have a marginally wider distribution of high achievers, though the core performance remains equivalent.

4. CORRELATION ANALYSIS

The correlation analysis reveals that attendance ($r = 0.58$) and hours studied ($r = 0.45$) have the strongest positive relationships with exam performance. Previous scores show a weaker but still positive association ($r = 0.18$). Other variables, such as sleep hours and physical activity, exhibit negligible relationships with exam outcomes. Additionally, the low correlations among independent variables suggest no multicollinearity concerns, supporting the reliability of subsequent regression analysis.

Table 4: Correlation Matrix

Variable	1	2	3	4	5	6	7
1. Hours Studied	1.00	—	—	—	—	—	—
2. Attendance	-0.01	1.00	—	—	—	—	—
3. Sleep Hours	0.01	-0.02	1.00	—	—	—	—
4. Previous Scores	0.03	-0.02	-0.02	1.00	—	—	—
5. Tutoring Sessions	-0.01	0.01	-0.01	-0.01	1.00	—	—
6. Physical Activity	0.00	-0.02	0.00	-0.01	0.02	1.00	—
7. Exam Score	0.45	0.58	-0.02	0.18	0.16	0.03	1.00

Note. $N = 6,607$. Correlations with $|r| \geq .024$ are statistically significant at $p < .05$ (two-tailed). Upper triangle replaced with em-dashes due to symmetry. Bold values indicate moderate-to-strong correlations with Exam Score. 1 = Hours Studied, 2 = Attendance, 3 = Sleep Hours, 4 = Previous Scores, 5 = Tutoring Sessions, 6 = Physical Activity, 7 = Exam Score.

The correlation matrix confirms that academic engagement variables (attendance and hours studied) are the primary correlates of exam success. The absence of high inter-correlations among predictors validates the assumption of no multicollinearity, ensuring that each predictor contributes unique variance in the regression models.

5. LINEAR REGRESSION ANALYSIS

A multiple linear regression was conducted with Exam Score as the dependent variable and Hours Studied, Attendance, Sleep Hours, Previous Scores, Motivation Level, Internet Access, Family Income, and Teacher Quality as independent variables. The goal was to quantify each factor's unique contribution to exam performance while controlling for all other variables.

Model Summary: $R^2 = 0.60$, Adjusted $R^2 = 0.60$, $F(12, 6594) = 829.40$, $p < .001$. The model explains approximately 60% of the variation in exam scores, suggesting a strong fit.

Table 5: Linear Regression Results

Predictor	B	SE	t	p
(Intercept)	42.34	0.44	95.47	< .001
Hours Studied	0.29	0.01	57.57	< .001
Attendance	0.20	0.00	76.04	< .001
Sleep Hours	-0.02	0.02	-1.03	.302
Previous Scores	0.05	0.00	22.68	< .001
Motivation Level (Low)	-1.03	0.09	-11.69	< .001
Motivation Level (Medium)	-0.49	0.08	-6.14	< .001
Internet Access (Yes)	0.85	0.11	7.38	< .001
Family Income (Low)	-1.09	0.08	-13.04	< .001
Family Income (Medium)	-0.58	0.08	-6.93	< .001
Teacher Quality (High)	0.59	0.28	2.07	.038
Teacher Quality (Low)	-0.46	0.29	-1.57	.117
Teacher Quality (Medium)	0.05	0.28	0.16	.870

Note. $N = 6,607$. Reference categories: Motivation Level (High), Family Income (High), Teacher Quality (High), Internet Access (No). $R^2 = .60$, Adjusted $R^2 = .60$, $F(12, 6594) = 829.40$, $p < .001$. Sleep Hours, Teacher Quality (Low), and Teacher Quality (Medium) were not statistically significant predictors.

Key Findings:

- $R^2 = .60$: The model explains 60% of variance in exam scores — strong predictive power.

- Attendance ($B = 0.20$): Every 1% increase in attendance $\rightarrow +0.20$ points exam score (strongest numeric predictor).
- Hours Studied ($B = 0.29$): Every additional hour studied $\rightarrow +0.29$ points (largest coefficient).
- Sleep Hours ($p = .302$): Not significant — sleep duration has no independent effect after controlling for other factors.
- Motivation & Income: Low motivation and low income significantly reduce scores relative to high categories.
- Teacher Quality: Only 'High' vs. reference shows weak significance ($p = .038$); Low/Medium are not significant.

Interpretation: Attendance and hours studied are the dominant predictors — together with previous scores, they drive most of the model's explanatory power ($R^2 = .60$), while sleep hours and most teacher quality levels show no significant independent effect.

Socioeconomic and motivational factors matter substantially: students with low motivation score ~ 1 point lower, low-income students score ~ 1.1 points lower, and having internet access provides a 0.85-point advantage, all net of study time and attendance.

6. LASSO REGRESSION RESULTS

Lasso regression was employed to perform variable selection by shrinking less important coefficients to zero. This approach provides a more parsimonious model, improving interpretability and reducing the risk of overfitting compared to standard ordinary least squares regression.

Table 6: Lasso Regression Results

Variable	Coefficient
Attendance	2.291
Hours Studied	1.762
Previous Scores	0.700
Tutoring Sessions	0.608
Internet Access	0.242
Motivation (Low)	-0.473
Family Income (Low)	-0.520

Note: Variables not listed were shrunk to zero and excluded from the model.

The Lasso regression results confirm that attendance, hours studied, and previous scores remain the most important predictors of academic performance. Additionally, tutoring sessions and internet access contribute positively, while lower motivation and lower family income negatively affect outcomes. Compared to the linear regression model, Lasso provides a more parsimonious model by excluding irrelevant variables, thereby improving interpretability and reducing the risk of overfitting. The consistency between OLS and Lasso results strengthens confidence in the robustness of the findings.

7. CLUSTER ANALYSIS

A k-means cluster analysis was conducted to identify natural groupings of students based on their academic profiles. Three distinct clusters were identified, each representing a unique combination of exam performance, study effort, and attendance behavior.

Table 7: Cluster Analysis Summary

Cluster	Exam Score	Hours Studied	Attendance
1 (High Performers)	70.92	24.11	86.45
2 (Low Performers)	64.31	18.62	69.32
3 (Average Performers)	67.61	17.71	88.25

Cluster Profiles:

- Cluster 1 (High Performers): Represents students with the highest exam scores (70.92), study hours (24.11), and attendance levels (86.45). These students demonstrate strong academic engagement across all dimensions.
- Cluster 2 (Low Performers): Consists of students characterized by lower attendance (69.32) and study effort (18.62), resulting in the lowest exam scores (64.31). This group may benefit from targeted intervention.
- Cluster 3 (Average Performers): Reflects students who maintain high attendance (88.25) but with lower study hours (17.71) and prior performance. Their exam scores (67.61) are moderate, suggesting that attendance alone is insufficient for top performance without adequate study effort.

These findings reinforce the regression results, highlighting the importance of academic engagement — particularly the combination of high attendance and substantial study hours — in determining student performance. The cluster analysis adds a practical dimension by identifying student segments that could inform differentiated educational interventions.

8. KEY FINDINGS AND CONCLUSIONS

This comprehensive analysis of 6,607 student records yields several important conclusions regarding the determinants of academic performance:

1. Academic Engagement is Paramount

Attendance and hours studied emerge as the strongest and most consistent predictors of exam success across all analytical approaches. Both correlation and regression analyses confirm that these behavioral variables have the largest effect sizes and remain significant even after controlling for socioeconomic and motivational factors.

2. Prior Achievement Matters

Previous scores demonstrate a positive and significant relationship with current exam performance, suggesting that academic success is partially cumulative. However, the effect size is smaller than that of current engagement behaviors, indicating that sustained effort remains more influential than past performance alone.

3. Socioeconomic and Motivational Disparities are Significant

Students from low-income families and those with low motivation levels score approximately 1 point lower on exams, even after controlling for study time and attendance. Internet access provides a modest but significant advantage (+0.85 points). These findings underscore the importance of addressing structural and motivational barriers to equity in educational outcomes.

4. Sleep and Teacher Quality Show Limited Independent Effects

Contrary to common assumptions, sleep hours do not exhibit a statistically significant independent effect on exam scores in this model. Similarly, only high teacher quality shows a marginal significant effect ($p = .038$), while low and medium quality levels are not significant. This suggests that student-level behaviors may be more predictive than environmental factors in this dataset.

5. Score Compression Limits Differentiation

The severe concentration of exam scores in the 60–70 range (81.4% of students) raises questions about the discriminatory power of the assessment instrument. This compression may attenuate predictive relationships and suggests that the exam may not adequately differentiate among higher-performing students.

6. Student Segmentation Reveals Actionable Profiles

The cluster analysis identifies three distinct student profiles that could inform targeted interventions: high performers who excel across all dimensions, low performers who need support in both attendance and study habits, and average performers who attend regularly but require additional study support to reach their potential.

Overall, this analysis demonstrates that student performance is primarily driven by controllable behavioral factors — attendance and study hours — rather than fixed characteristics. While socioeconomic background and motivation play meaningful roles, the prominence of effort-related variables suggests that targeted educational interventions focused on increasing academic engagement could yield substantial improvements in student outcomes. Future research should explore the causal mechanisms underlying these relationships and investigate whether the observed score compression reflects a genuine ceiling effect or a need for more discriminatory assessment design.